Name: Cian Mac Liatháin **ID:** 11476078

ED305 Assignment 1: Inclusive Teaching Paper

A. Context:

This lesson plan is designed based on the context of the school I undertook teaching practice in this year. The school is mixed-sex, in an urban setting, and caters for roughly 300 post-primary students. The school is classified as a DEIS school, and the student population consists of diverse socio-economic and cultural backgrounds, including minority ethnic backgrounds as well as members of the Travelling community. Some students have had difficulty in other schools previously and have now found a place in this school which prides itself in providing an education for all types of students based on their individual strengths, emphasising an interdenominational, academic and vocational ethos. The school also provides education in the form of post-Leaving Cert PLC courses and part-time adult evening courses, which are beneficial to all types of students as alternative pathways to higher education.

The class group is a Transition Year maths class of 20 students. As a Transition Year group, they are a mixture of both higher level and ordinary level maths students. The students are primarily from low socio-economic backgrounds and are culturally diverse. The group includes 5 visiting Spanish students who have English as an Additional Language, as well as 2 non-Traveller minority ethnic students.

This lesson will be their first lesson in Statistics since completing the Junior Certificate the year before. For the students visiting from Spain, they have previously visited the same statistical topics during their time studying for the Spanish equivalent of the Junior Certificate, *Educación Secundaria Obligatoria.* All students have previously encountered the ideas of data collection and statistical analysis, and in this lesson they will be refreshing their knowledge by carrying out their own statistical investigation. This will provide them with a solid statistical foundation before carrying out further statistical investigations in later classes.

Nearly all students in this class exhibit poor engagement with traditional methods of teaching mathematics through repeated practice of questions and rote learning. Their disengagement leads to a lack of communication in the class, as well as difficult to control behaviour. Furthermore, the visiting students from Spain only talk to each other, if at all. Similarly, the two ethnic-minority students only sit next to and interact with each other. The visiting

students frequently refuse to answer questions asked of them, simply stating that they 'can't do it'. None of the students are expected to have a textbook, and some normally don't have a pen or a copy, due to difficulties affording them. In previous classes, two female students expressed a dislike of mathematics and dismissed any discussion about its usefulness, commenting "I can't go to college anyway, I'll only be a hairdresser".

In my opinion, inclusive teaching in the context of this class involves making the topic of mathematics more accessible to all students, in an effort to promote communication and participation in the class. My own teaching and communication should be done in a way that is conscious of students with English as an additional language. The lesson should be designed in a way that promotes peer learning and communication between all types of students, as well as facilitates students that aren't coming into the class with their own materials. I hope to encourage students towards mathematics by highlighting its relevance to all professions and dismissing myths about 'not being a maths person'. My aim is to ensure that neither language, culture nor gender is a barrier to the mathematical learning outcomes.

B. <u>Inclusive Lesson Plan:</u>

Highlighting the diversity of students' personalities, experiences, and viewpoints, Tomlinson (2017) emphasises the importance of teachers carrying out differentiated instruction in their classrooms. A differentiated classroom "provides different avenues to acquiring content, to processing or making sense of ideas, and to developing products so that each student can learn effectively" (Tomlinson, 2017, p.1). In order for me to create an inclusive lesson, I feel it is important to carefully consider not only the design of the lesson itself but also the principles that underly my teaching. Tomlinson (2017) outlines principles that "can be helpful in ensuring that struggling learners maximise their capacity in school", and from these I have identified those that are important to me as a pre-service teacher:

Teach with a growth mindset and with a belief of the capacity of each child – this is a principle that I feel is very important, as it avoids placing blame at the feet of the students for difficulties in learning. Believing that all the students in my class have an inherent capacity for growth and learning allows me to focus on improving my own teaching to meet their needs. For example, when the EAL students sometimes declare their inability to do any maths, I won't simply agree with them and focus on other 'easier' students. By possessing a belief in their capacity for growth and learning, their feelings of inability give me valuable insight, and motivate me to identify challenges from their point of view. Perhaps my English

speaking is too fast, or my Irish accent is too strong. These are easily fixable problems as long as I remain conscious of them, which I hope to do as I carry out the lesson outlined below. Tomlinson (2017, p.27) advises that working to ensure EAL students feel acknowledged and accepted can greatly improve their learning and participation in class, and I hope to do this by consistently offering opportunities to talk, as well as integrating Spanish key vocabulary lists into the lesson to "build bridges between students' first languages and English".

Pay attention to relevance – this is an important principle that I feel will help all types of students in the class. In different ways, many students fail to see the relevance of any maths topic approached in a traditional way that doesn't acknowledge the diversity of the students in the class. Students who don't think they have any mathematical potential grow disengages in a similar way to the EAL students, after too much abstract jargon is used in my teaching. This lesson is designed in a way that emphasises the relevance to the students' lives, and will hopefully work as a foundation in making future mathematics lessons accessible as well.

Learning Outcomes

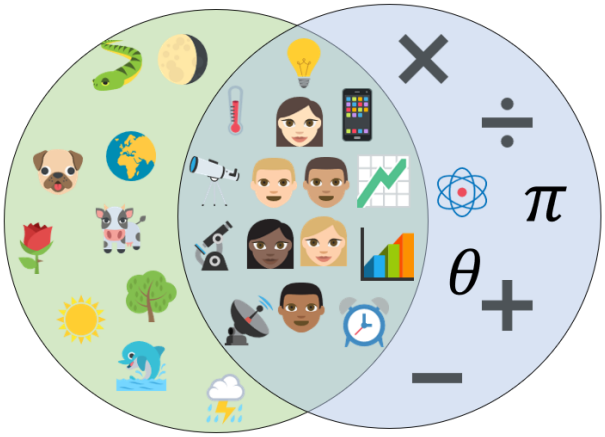
One of the main learning outcomes for this lesson is for the students to develop a statistical view of the world and grow more statistically aware. I hope for students to develop an understanding of the importance and relevance of mathematical thinking to everyday life.

By the end of the lesson, students should be able to understand what is meant by the Data Handling Cycle. When explained with worldwide examples, I expect that students will be able to recognise statistical investigations, data collection and its applications already present in their everyday lives.

Carrying out their own statistical investigation, students will work in pairs to practice and know how to mathematically calculate the Mean, Median and Mode of data they collect, and through discussion with other pair-groups discuss their usefulness.

Lesson Layout

The lesson will be opened with a PowerPoint slide of a Venn diagram, with one side containing images/symbols related to nature, another with symbols related to mathematics and the centre with symbols of humans and examples of how we apply the right side to the left. This will be used as a hook, to introduce mathematical, statistical and scientific thinking in the context of our shared daily experiences. Example of this given below:



The use of open questioning has been described by Small (2012, p.6) of Mathematics as a "core strategy" of differentiating instruction in a mathematics classroom. Open questioning allows for teachers to encourage the expression of different points of view and promotes confidence in students (Small, 2012) This will be used to prompt discussion in the class about the students' opinions of mathematics and its applications. Example questions include: "Do you think maths is useful? What are uses of maths? Has maths changed our lives? How does maths make you feel?". The teacher can describe the centre of the Venn diagram by giving examples of how we use maths to make sense of the world, how it is prevalent in our everyday lives, and giving examples of how different professions use maths. Maths is framed as a universal language.

After engaging with the students, the lesson continue on with the concept of a Venn diagram, asking "Why do we say Pythagoras' Theorem? Why is this called a Venn diagram?". The NCCA (2006, p.70) has identified that exploring "the contribution of diverse cultures to our mathematical culture" is a key aspect of an inclusive Mathematics programme. The teacher will describe mathematics as a joint effort, featuring contributions from many different types of people and dispelling the myth of a singular type of 'maths person'. The teacher will show the photo of, and give brief descriptions of a variety of people who have contribute to the field of mathematics. For example:

John Venn – a male English mathemetician famous for contributions to statistics, including the Venn diagram.

Hypatia – a famous female mathematician from Alexandria, Egypt, c.350-370 AD. Also a philosopher and astronomer.

Yutaka Taniyama – a male Chinese mathematician significant contributions to modern geometry that we use today.

Maria Assumpcio Cata i Poch – a female Spanish mathematician contributions to mathematical analysis, important work involved in calculating orbits.

David Blackwell – a male African-American mathematician significant contributions to statistics and probability.

Florence Nightingale – a female English mathematician who made significant improvements to data collection and using it to improve health services.

The lesson will continue by asking students about the use of data in modern society. As all students have smartphones, the concept will be introduced in terms of asking how companies like Apple, Google, Amazon, Facebook collect and use data. Pictures of two different Amazon reviews, in Spanish and English, will be used to discuss quantitative and qualitative data, their uses, and the accuracy of data and statistics (e.g. an item with a single wordless 5-star rating vs an item with thirty 4.5-star reviews with detailed descriptions).

According to the NCCA (2006, p.69), presenting opportunities for students to "examine information on local and global issues" is an important aspect of an inclusive Mathematics programme in terms of intercultural education. The teacher will present the interactive COVID-19 map organized by John Hopkins University (2020). This will be used as an example of the shared community of mathematics, asking questions like: " Why should we be concerned about global data? How does this data from other countries help us here in Ireland? Is it important for countries to communicate with each other?". This resource is particularly useful for prompting discussion on statistical accuracy and data collection, as there are interactive statistics available that allow the teacher to discuss why infection rates could be lower in countries that are simply testing less, for example.

Another important aspect of an inclusive Mathematics programme, according to the NCCA (2006, p.69) is giving students opportunities to "engage in group activities and investigative learning". Tomlinson (2017) advises that there are different requirements for effective group work to take place. The goals of the task should be made clear to the students, and groups should be designed based on "group members' strengths and interests" as well as in a way that "requires group collaboration to achieve shared understanding and quality outcomes (Tomlinson, 2017, p.48). Based on this, the members of each group-pair will be chosen by

the teacher in order to promote the collaboration and communication of the students between different class subgroups. This investigation is designed in a way that promotes communication between the two members of each group in addition to communication between the groups themselves. This is guided by principles of co-operative learning (Clarke, 1994)

The students will investigate the contents of Smarties tubes provided to each pair of students by the teacher, guided by the worksheet attached in the Appendix below. The teacher will also provide colouring pencils and pens to students without any. The investigation is based around the fact that Smarties tubes don't give specific numbers for their contents, instead using an approximation per tube. The students will investigate, counting the Smarties, calculating the Mean/Median/Mode and graphing the data. After the initial investigation carried out by pairs of students, one member of each pair will travel to students that investigated in other pairs. The class as a whole will compare their data to each other and will try to get a whole-class perspective.

The investigation is intended to promote discussion between all students of the class including EAL students and ethnic-minority students. From the beginning of the lesson the teacher will have been contributing to a bilingual word key, and students are encouraged to communicate in any way they feel comfortable. As it is being filled, the teacher will attempt to frame the foreign language and other cultures in a positive light. For example, it could be explained that many languages are closely related and all languages are useful. An example discussion point could be, which word has a more intuitive meaning, 'Accuracy' or 'La exactidud'? Example word key:

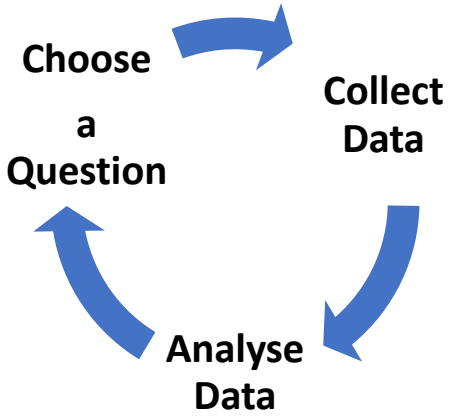
English	**Spanish**
Data	Datos
Collect	Recoger
Analyse	Analizar
Interpret	Interpretar
Investigation	Investigación
Statistics	La Estadística
Accuracy	La exactitud
Qualitative	Cuantitativa
Quantitative	Cualitativa
Graphs	Gráfico
Bar chart	Gráfico de barras
Measure	Medida
Mean	Media
Median	Mediana
Mode	Moda
Range	Rango

After the students carry out their investigation, communicating in pair-groups and between pair-groups, the teacher will congratulate them on carrying out a successful statistical investigation. An emphasis will be placed on the importance of collaboration, and the teacher will explain how future lessons will continue to build on their statistical reasoning and co-operation. A few examples of Census@School (2020) data and statistics will be shown from international classrooms, and it will be explained that we will carry out similar investigations in the coming classes.

Smarties Investigation

Try not to open them until Part 2! 😉

Aim: To develop our statistical knowledge through investigation, using the Data Handling Cycle:



Part 1 - Choose a Question:

- Today we'll investigate the contents of a tube of Smarties. Can we find any information on the tube telling us *how many Smarties are inside?*

_____

- Based on the picture on the tube, *how many different colours should there be?*

_____

- Can you come up with a question that describes what we want to investigate?

_____

Part 2 - Collect Data:

- Pour the Smarties out and find the Total Amount: _____
- Sort them in columns of each colour.
- Gather data for each colour:

Colour										
Amount										

Part 3 - Analyse Data:

- How many different colours are there? _____
- What is the **Mean** amount of Smarties?

- What is the **Median** amount?

- What is the **Mode?**

- Calculate the **Range** of the amounts of Smarties:

- Are these the same as the results of the person next to you?

- Draw a **Bar chart** of your Data**:**

Bibliography

Census at School (2020) Available at: https://censusatschool.ie/

National Council for Curriculum and Assessment (2006) *Intercultural Education in the Post-Primary Schools: Guidelines for Schools.* (Accessed at: https://developmenteducation.ie/media/documents/IntercEdPostPrimaryGuide.pdf)

Smart, M. (2012) *Good Questions: Great Ways to Differentiate Mathematics Instruction.* Second Edition. New York, USA: Teachers College Press.

Tomlinson, C. (2017) *How to Differentiate Instruction in Academically Diverse Classrooms.* Third Edition. Virginia, USA: ASCD.